

DNP3 driver

This module is a master (client) driver for DNP3 (IEEE 1815), the polled protocol common across North American electric distribution and water/wastewater. It reads only; it does not control.

Version 0.1.0 Ignition 8.1.33+ Runs on: any gateway that reaches the outstation network

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1. Overview

The driver is a standalone OPC UA device type; it does not require any other SparkBridge module. TCP transport only; serial is not implemented.

2. What You Need

- Ignition 8.1.33 or newer. TCP transport only.
- An outstation reachable over TCP, registered port 20000.

3. Install

Install the signed DNP3 module through Config > Modules > Install or Upgrade a Module; no build step is required. Module id `com.greenguard.drivers.dnp3`.

4. License

See the README in the bundle for licensing terms.

From release 2.7.0, every SparkBridge module runs on Ignition's standard two-hour module trial (the gateway's Reset Trial button restarts the clock). A verified key overrides the trial entirely and the trial clock is then ignored. Drop the key at `<data>/sparkbridge-license.key`; each module re-checks the file every 30 seconds, so installing, replacing, or removing a key takes effect with no restart.

5. Configure

Config > OPC UA > Device Connections > Create new Device > DNP3.

Setting	Meaning, default
Hostname/IP	Outstation address.
Port	TCP port, registered default 20000.
Master Address	This driver's own DNP3 link address.
Outstation Address	The outstation's DNP3 link address, per station.

Addressing is type plus index, per type: BI7 (binary input 7), DB3 (double-bit binary input 3), CTR0 (counter 0), AI12 (analog input 12). Group-number spellings are also accepted, for example G1.7 or G30.12. No type suffix is accepted; the outstation decides the data type.

6. Operate

Addresses resolve on demand and answer `Bad_WaitingForInitialData` until first reported. Browse shows four folders (BI, DB, CTR, AI) built from the integrity poll; the browse tree is empty until the first integrity poll completes, which is expected behavior on a fresh device, not a fault.

7. Verify

- Run the bundled probe: `java -cp common/build/libs/common-0.1.0.jar com.greenguard.dnp3.tools.Dnp3Probe --host <ip> --port 20000 --master 1 --outstation 10 --seconds 60.`
- Confirm the device reaches `READY` and the status line reports a nonzero point count after the first integrity poll.

8. Troubleshoot

Symptom	Cause / fix
Browse tree is empty	Normal until the first integrity poll completes; wait for it or check that Master Address and Outstation Address match the outstation's configuration.
Points read Bad_WaitingForInitialData	Expected until the point is first reported by the outstation; not itself an error.
Writes are rejected	Expected: this driver reads only and does not implement control operations.

9. Reference

Module id	<code>com.greenguard.drivers.dnp3</code>
Default port	TCP 20000
Browse folders	BI, DB, CTR, AI
Capability	Read only, no control

10. Support

Docs page section: </sparkbridge/docs#dnp3>. Contact: </sparkbridge/contact>. Release downloads and checksums: [GitHub releases](#).